

Estimation Multivariate Normal Algorithm with Threshold Convergence

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Abstract:

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Kullback-Leibler Divergence

The **Kullback-Leibler (KL) divergence**, denoted by $D_{\text{KL}}(P \parallel Q)$, measures how much an approximating probability distribution Q is different from a true probability distribution P . Mathematically, it is defined as

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in X} P(x) \log \left(\frac{P(x)}{Q(x)} \right).$$

One may also extend the definition to continuous random variables as

$$D_{\text{KL}}(P \parallel Q) = \int_{\mathbb{R}} p(x) \log \left(\frac{p(x)}{q(x)} \right) dx$$

An interpretation of the KL divergence of P from Q is the expected excess surprisal from using the approximation Q instead of P when the actual is P . It is important to note that this is not a metric, as in general, the symmetric property $D_{\text{KL}}(P \parallel Q) = D_{\text{KL}}(Q \parallel P)$ does not hold (neither does the triangle inequality). It does satisfy the properties $D_{\text{KL}}(P \parallel P) = 0$ and $D_{\text{KL}}(P \parallel Q) \geq 0$.

KL-Divergence of Normal Distributions

A particularly interesting application of the KL divergence is to two normal distributions, which may pop-up in almost any real-life phenomenon by the central limit theorem. This can help us gauge how much information is lost when attempting to approximate one normal distribution with another.

Let $P = \mathcal{N}(\mu_1, \sigma_1^2)$ and $Q = \mathcal{N}(\mu_2, \sigma_2^2)$. We first note that

$$\begin{aligned} D_{\text{KL}}(P \parallel Q) &= \int_{\mathbb{R}} p(x) \log \left(\frac{p(x)}{q(x)} \right) dx \\ &= \int_{\mathbb{R}} p(x) \log p(x) dx - \int_{\mathbb{R}} p(x) \log q(x) dx. \end{aligned}$$

The first term gives

$$\begin{aligned}
\int_{\mathbb{R}} p(x) \log p(x) \, dx &= \int_{\mathbb{R}} \left(-\frac{1}{2} \log(2\pi\sigma_1^2) - \frac{(x - \mu_1)^2}{2\sigma_1^2} \right) p(x) \, dx \\
&= -\frac{1}{2} \log(2\pi\sigma_1^2) \underbrace{\int_{\mathbb{R}} p(x) \, dx}_1 - \frac{1}{2\sigma_1^2} \underbrace{\int_{\mathbb{R}} (x - \mu_1)^2 p(x) \, dx}_{\sigma_1^2} \\
&= -\frac{1}{2} \log(2\pi\sigma_1^2) - \frac{1}{2}.
\end{aligned}$$

For the second term, we have

$$\begin{aligned}
\int_{\mathbb{R}} p(x) \log q(x) \, dx &= \int_{\mathbb{R}} \left(-\frac{1}{2} \log(2\pi\sigma_2^2) - \frac{(x - \mu_2)^2}{2\sigma_2^2} \right) p(x) \, dx \\
&= -\frac{1}{2} \log(2\pi\sigma_2^2) \int_{\mathbb{R}} p(x) \, dx - \frac{1}{2\sigma_2^2} \int_{\mathbb{R}} (x - \mu_2)^2 p(x) \, dx \\
&= -\frac{1}{2} \log(2\pi\sigma_2^2) - \frac{1}{2\sigma_2^2} \left[\int_{\mathbb{R}} x^2 p(x) - 2\mu_2 x p(x) + \mu_2^2 p(x) \, dx \right]
\end{aligned}$$

We now use the notation $\mathbb{E}_p[g(x)] = \int g(x)p(x) \, dx$, the expected value of $g(x)$ under $p(x)$. We note that $\mathbb{E}_p[x^2] - \mathbb{E}_p[x]^2 = \text{Var}(x)$, and thus $\mathbb{E}_p[x^2] = \sigma_1^2 + \mu_1^2$. Rewriting the last expression gives

$$\begin{aligned}
\int_{\mathbb{R}} p(x) \log p(x) \, dx &= -\frac{1}{2} \log(2\pi\sigma_2^2) - \frac{1}{2\sigma_2^2} \left[\int_{\mathbb{R}} x^2 p(x) - 2\mu_2 x p(x) - \mu_2^2 p(x) \, dx \right] \\
&= -\frac{1}{2} \log(2\pi\sigma_2^2) - \frac{1}{2\sigma_2^2} [\mathbb{E}_p[x^2] - 2\mu_2 \mathbb{E}_p[x] + \mu_2^2] \\
&= -\frac{1}{2} \log(2\pi\sigma_2^2) - \frac{1}{2\sigma_2^2} [\sigma_1^2 + \mu_1^2 - 2\mu_2 \mu_1 + \mu_2^2] \\
&= -\frac{1}{2} \log(2\pi\sigma_2^2) - \frac{1}{2\sigma_2^2} [\sigma_1^2 + (\mu_1 - \mu_2)^2]
\end{aligned}$$

Putting everything together, we get

$$\begin{aligned}
D_{\text{KL}}(P \parallel Q) &= \frac{1}{2} \log \left(\frac{\sigma_2^2}{\sigma_1^2} \right) + \frac{\sigma_1^2}{2\sigma_2^2} + \frac{(\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2} \\
&= \log \left(\frac{\sigma_2}{\sigma_1} \right) + \frac{\sigma_1^2}{2\sigma_2^2} + \frac{(\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}.
\end{aligned}$$

In particular, if $\sigma_1 = \sigma_2$, we have

$$D_{\text{KL}}(P \parallel Q) = \frac{(\mu_1 - \mu_2)^2}{2\sigma_2^2} = D_{\text{KL}}(Q \parallel L).$$

We may also note that taking the limit as $\sigma_1 \rightarrow 0^+$ (and fixing σ_2) yields an infinite value (since $\frac{1}{x^2}$ grows faster than $\log x$ as $x \rightarrow 0^+$), and the same can be said for $\sigma_2 \rightarrow 0^+$ (and fixing σ_1). This can be interpreted as it being equally as surprising if we obtain information when none was emitted, as it is to not obtain any when it was sent.

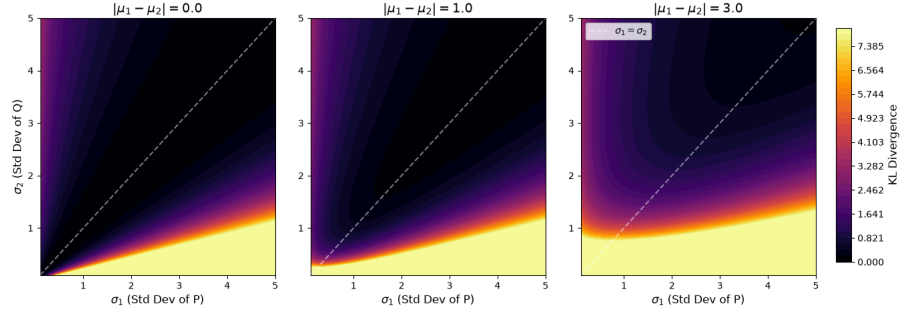


Figure 1: Level curves of the KL divergence for some values of $|\mu_1 - \mu_2|$. We note the **asymmetry** seen in the equation.

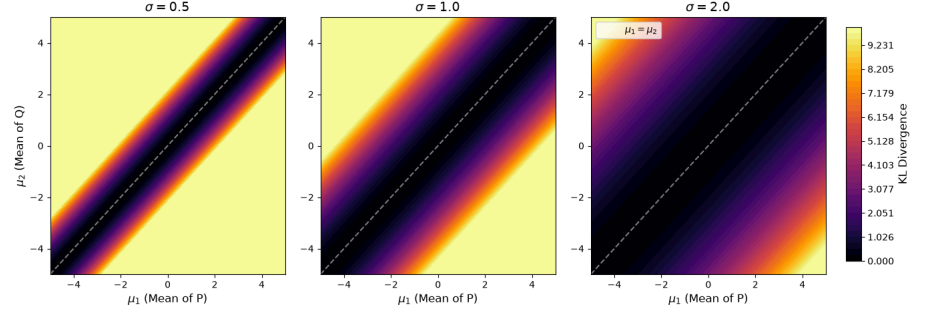


Figure 2: Level curves of the KL divergence for some values of $\sigma_1 = \sigma_2 = \sigma$. We note the **symmetry** seen in the equation.

Mutual Information

The **mutual information** of two random variables is a measure of the mutual dependence between the two variables. It quantifies the “amount of information” obtained about one random variable by observing the other random variable.

Let (X, Y) be a pair of random variables with values over the space $\mathcal{X} \times \mathcal{Y}$. If their distribution is $P_{X,Y}$ and the marginal distributions are P_X and P_Y , the mutual information is defined as

$$I(X; Y) = D_{\text{KL}}(P_{X,Y} \parallel P_X \times P_Y)$$

where $P_X \times P_Y$ is the outer product distribution which assigns probability $P_X(x) \cdot P_Y(y)$ to each (x, y) .

Mutual Information of two Normal Distributions

Let $X = \mathcal{N}(\mu_X, \sigma_X)$ and $Y = \mathcal{N}(\mu_Y, \sigma_Y)$. Then, we recall the following:

$$p(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{1-\rho^2} \left[\frac{(x-\mu_X)^2}{\sigma_X^2} - 2\rho \frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right]\right)$$

We define $\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} =: \kappa$ to shorten the expression.

Now, by definition of the KL divergence,

$$\begin{aligned} I(X; Y) &= D_{\text{KL}}(P_{X,Y} \parallel P_X \times P_Y) \\ &= \int_{\mathbb{R}^2} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx \\ &= \int_{\mathbb{R}^2} p(x, y) \log p(x, y) dx dy - \int_{\mathbb{R}^2} p(x, y) \log p(x) dx dy - \int_{\mathbb{R}^2} p(x, y) \log p(y) dx dy \\ &= \mathbb{E}[\log p(x, y)] - \mathbb{E}[\log p(x)] - \mathbb{E}[\log p(y)] \end{aligned}$$

We then have that

$$\begin{aligned}\mathbb{E}[\log p(x, y)] &= \log \kappa - \frac{1}{2(1-\rho^2)} \left[\frac{\mathbb{E}[(x - \mu_X)^2]}{\sigma_X^2} - \frac{2\rho}{\sigma_X \sigma_Y} (\rho \sigma_X \sigma_Y) + \frac{\sigma_Y^2}{\sigma_X^2} \right] \\ &= \log \kappa - \frac{1}{2(1-\rho^2)} [1 - 2\rho^2 + 1] \\ &= \log \kappa - 1.\end{aligned}$$

The other terms simplify in the following manner:

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} p(x, y) \, dy \right) \log p(x) \, dx = \int_{\mathbb{R}} p(x) \log p(x) \, dx,$$

which we calculated earlier to be

$$-\frac{1}{2} \log(2\pi\sigma_1^2) - \frac{1}{2}.$$

Similarly,

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} p(x, y) \, dx \right) \log p(y) \, dy = \int_{\mathbb{R}} p(y) \log p(y) \, dy$$

evaluates to

$$-\frac{1}{2} \log(2\pi\sigma_2^2) - \frac{1}{2}.$$

Altogether, we have

$$I(X; Y) = \log \kappa + \log(2\pi\sigma_1\sigma_2) = \log \frac{2\pi\sigma_X\sigma_Y}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} = -\frac{1}{2} \log(1-\rho^2).$$

In particular, we notice that if $\rho = 0$ (the variables have no correlation),

$$I(X; Y) = 0.$$

If $\rho \rightarrow 1^-$,

$$I(X; Y) \rightarrow \infty.$$

And similarly, if $\rho \rightarrow -1^+$,

$$I(X; Y) \rightarrow \infty.$$

We also note that the mutual information does not depend on the means of each distribution. This is due to the entropy only caring about the variance of the data, and not the locality.